

Browser Fingerprinting Lab

Passive, Active, Typing and Hash Features

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Objective

Observe information sent automatically by the browser in an HTTP request, collect selected headers with Flask, and compare a normal window with a private browsing window. No JavaScript is needed for this part.

Method

I ran the local Flask application at `localhost:8000`. In Developer Tools, I inspected **Network** → **localhost** → **Headers** → **Request headers**. I then printed selected values on the server using `request.headers.get(...)` and repeated the visit in private browsing mode.

Passive collector

The route printed the client IP, HTTP method, **User-Agent**, and **Accept-Language**. These lines collected additional request headers:

```
print("Host:", request.headers.get("Host"))
print("Accept:", request.headers.get("Accept"))
print("Sec-Fetch-Mode:", request.headers.get("Sec-Fetch-Mode"))
print("Sec-CH-UA-Platform:", request.headers.get("Sec-CH-UA-Platform"))
print("Sec-Fetch-Site:", request.headers.get("Sec-Fetch-Site"))
```

The terminal also printed `From`, which was absent (`None`) in both requests.

Observed results

Feature	Value in both windows	Meaning
IP / method	127.0.0.1 / GET	Local client; page request
User-Agent	Windows NT 10.0; Chrome/154.0.0.0	Browser and OS string
Accept-Language	fr-FR,fr;q=0.6	Preferred languages
Host	localhost:8000	Requested server
Accept	HTML, XHTML, XML, image formats, etc.	Accepted response formats
Sec-Fetch-Mode	navigate	Page navigation
Sec-CH-UA-Platform	"Windows"	Reported platform
Sec-Fetch-Site	none	Direct user navigation
From	None	Header not sent

Conclusion

In this test, **none of the collected values changed** between the normal and private windows. Private browsing did not hide these HTTP request properties. These shared values alone do not prove that the two visits came from a unique person or browser: other users may have similar values. The local IP `127.0.0.1` identifies this computer's loopback connection, not its public IP.

Evidence from the Flask terminal

Normal browsing window

```
===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : fr-FR,fr;q=0.6
Host : localhost:8000
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-Fetch-Mode : navigate
Sec-CH-UA-Platform: "Windows"
Sec-Fetch-Site : none
From : None
=====

127.0.0.1 - - [29/Sep/2026 12:16:54] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 12:16:54] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
█
```

Private browsing window

```
===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : fr-FR,fr;q=0.6
Host : localhost:8000
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-Fetch-Mode : navigate
Sec-CH-UA-Platform: "Windows"
Sec-Fetch-Site : none
From : None
=====

127.0.0.1 - - [29/Sep/2026 12:20:48] "GET / HTTP/1.1" 200 -
█
```

The screenshots record the values printed by the Flask server for the `GET /` request. The additional DevTools-related 404 visible after the normal visit is separate from the successful page request (200).

Part 2 – Expand the Active Collector

Objective. JavaScript read browser properties, displayed them on the page, and sent them to the Flask server with a JSON POST `/collect` request. Unlike Part 1, this required code running in the browser.

Feature and object	What it reveals	Possible fingerprint value / limitation
<code>navigator.language</code>	Preferred browser language	Narrows a group; common among many users.
<code>screen.availWidth</code>	Available screen width	May reflect display setup; can change with display or browser environment.
<code>window.innerWidth</code>	Current page viewport width	Varies when the window or DevTools is resized.
<code>Intl.time zone</code>	Regional time zone	Narrows a region; shared by many users.
<code>Intl.Locale</code>	Locale derived from browser language	Often duplicates language information here.
<code>document.visibility</code>	Whether this tab is visible	Describes current tab state, not a stable browser identity.

How the data reached Flask

The six values were placed in a JavaScript `features` object and displayed in the page and browser console. The client sent `JSON.stringify(features)` with `fetch('/collect', {method: 'POST', ...})` and `Content-Type: application/json`. The Flask `/collect` route read `request.get_json()` and printed each received value. The server screenshot shows POST `/collect` with status 200.

Comparison with private browsing

Feature	Normal window	Private window
Browser language	<code>fr-FR</code>	<code>fr-FR</code>
Screen available width	1280	1680
Window inner width	632	1506
Time zone	<code>Africa/Casablanca</code>	<code>Africa/Casablanca</code>
Locale	<code>fr-FR</code>	<code>fr-FR</code>
Visibility state	<code>visible</code>	<code>visible</code>

Interpretation. Four observed values stayed the same; two widths changed. The screenshots do not establish why the widths changed: viewport size, available display area, or display setup can differ between tests. Private mode alone did not remove the language and regional clues. Cookie blocking and privacy extensions were not tested in the supplied evidence, so this experiment cannot rank the protections or identify the most effective one. Fingerprinting can still use features that remain visible, and individual values are not proof of identity.

Part 2 – Evidence

Normal window: values on the page and in the console

that is my html for the lab :)

Active Feature Collection

Browser language: fr-FR
Screen available width: 1280
Window inner width: 632
Intl timezone: Africa/Casablanca
Intl locale: fr-FR
Document visibility state: visible

==== Active Feature Collection ====

Browser language: fr-FR
Screen available width: 1280
Window inner width: 632
Intl timezone: Africa/Casablanca
Intl locale: fr-FR
Document visibility state: visible

(index)	Value
Browser language	'fr-FR'
Screen available width	1280
Window inner width	632
Intl timezone	'Africa/Casablanca'
Intl locale	'fr-FR'
Document visibility state	'visible'

Object

Flask terminal: received values and successful POST

```
=== ACTIVE FEATURES RECEIVED ===
Browser language: fr-FR
Screen available width: 1280
Window inner width: 632
Intl timezone: Africa/Casablanca
Intl locale: fr-FR
Document visibility state: visible
127.0.0.1 - - [29/Sep/2026 17:25:28] "POST /collect HTTP/1.1" 200 -
```

Private window: received values

```
===== NEW VISIT =====
=====

127.0.0.1 - - [29/Sep/2026 17:28:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 17:28:34] "GET /static/fingerprint.js HTTP/1.1" 200 -

=== ACTIVE FEATURES RECEIVED ===
Browser language: fr-FR
Screen available width: 1680
Window inner width: 1506
Intl timezone: Africa/Casablanca
Intl locale: fr-FR
Document visibility state: visible
127.0.0.1 - - [29/Sep/2026 17:28:34] "POST /collect HTTP/1.1" 200 -
```

Task 3 – Typing Experiment Page

The webpage showed the exact target sentence, a text input, a Restart button and an initially empty results area. Each interactive element had its own ID so JavaScript could access it. The sentence was:

Cookies are gone, but Alice's rhythm still leaves traces!

The actual page used the curly apostrophe in *Alice's*; the typed text had to match the displayed sentence exactly.

Task 4 – Alice's Typing Rhythm

The JavaScript started the timer when the first character appeared, counted Backspace presses, and stopped when the input exactly matched the target sentence. Restart cleared the text, result and measurements. After completion, the browser displayed and sent the results as JSON to `/collect-typing`; Flask printed them.

Typing speed (characters per second):

$$\text{speed} = \frac{\text{characters in the target sentence}}{\text{time in seconds}}$$

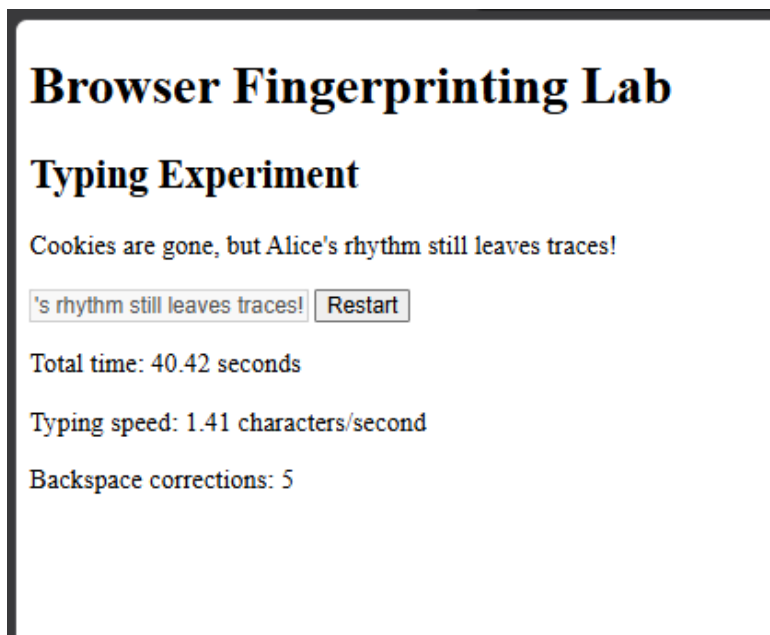
Spaces and punctuation count as characters.

Measurement	Observed value
Total typing time	40.42 seconds
Typing speed	1.41 characters/second
Backspace corrections	5
Server request	POST <code>/collect-typing</code> – 200

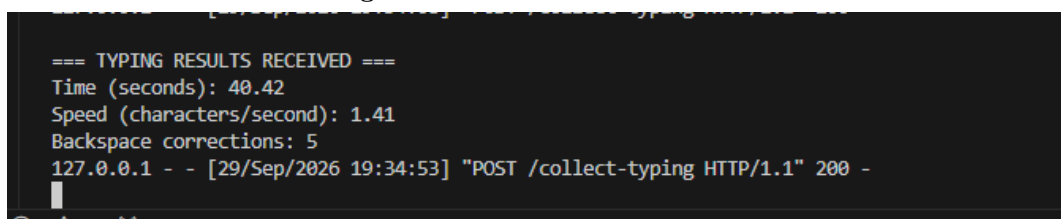
Observation. The browser and Flask terminal show the same three results. This is one typing sample; speed and corrections can vary on another attempt, so they are not a certain identifier of a person.

Evidence

Webpage after typing



Flask terminal after receiving the results



Part 4 – Give the Clues One Identifier (Optional)

The browser combined selected environment values in a fixed order and computed a SHA-256 hash. Typing time, speed and corrections were excluded. JavaScript displayed the hexadecimal fingerprint and sent it to `/collect-fingerprint`; Flask printed the received identifier.

language | screen width | time zone | locale → SHA-256 fingerprint

Observed result. The normal and private windows produced **different fingerprints**. Both Flask logs show a successful `POST /collect-fingerprint` (status 200). The earlier active-feature experiment also showed a different available screen width between these windows (1280 versus 1680), which could contribute to the changed hash if this feature was included. A hash changes whenever an input changes; the screenshots alone do not establish which input caused the difference.

Limit. Repeated visits in the same mode and comparisons with colleagues were not documented, so stability and uniqueness were not demonstrated. Hashing makes the raw features less immediately readable in this request, but the same hash can still be used to link visits. It does not prove a person's identity.

Evidence

Normal window: fingerprint received by Flask

```
===== NEW VISIT =====
=====

127.0.0.1 - - [29/Sep/2026 22:35:09] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 22:35:09] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 - - [29/Sep/2026 22:35:09] "GET /static/fingerprint.js HTTP/1.1" 304 -
127.0.0.1 - - [29/Sep/2026 22:35:09] "GET /static/fingerprint3.js HTTP/1.1" 304 -
Fingerprint: 2ad9d859bd8ccb0d059485922e985ebb40abc7290203868ae96f2e9ad24240c0
127.0.0.1 - - [29/Sep/2026 22:35:09] "POST /collect-fingerprint HTTP/1.1" 200 -
```

Private window: a different fingerprint received by Flask

```
===== NEW VISIT =====
=====

127.0.0.1 - - [29/Sep/2026 22:36:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 22:36:29] "GET /static/fingerprint3.js HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 22:36:29] "GET /static/fingerprint.js HTTP/1.1" 200 -
Fingerprint: 7c7be95fcb036be87cfe0276705fdb7c7f9618ea0472fa2fc500a15f6dc335ff
127.0.0.1 - - [29/Sep/2026 22:36:29] "POST /collect-fingerprint HTTP/1.1" 200 -
```